meesho SEASON 3 DICE CHALLENGE

MEESHO DICE CHALLENGE 3.0 • DETAILED SUBMISSION ROUND

PROJECT NIRMAN: ZERO-LEAKAGE LAST MILE

A Data-Driven Strategy to Reduce RTO & Monetize Refused Inventory on ValMo

TRACK: VALMO LOGISTICS (REDUCING RTO) • TEAM FurryCat32

KEY STRATEGIC DELIVERABLES & CORE PILLARS:

- Rigorous Quantitative Diagnosis: Unbundling the ₹3,485 Cr annual RTO bleed across 1.2B shipments using Meesho's explicit data pack.
- ValMo SmartAttempt™: AI Pre-Dispatch WhatsApp Concierge, Truecaller Verified Telephony & 150m Geo-Fence Anti-Ghosting Guard.
- ValMo Re-Route™: Hyperlocal re-commerce engine staging refused parcels at LMDC, saving 100% of the ₹120 reverse freight.
- Primary Field Research: Grounded in 15 gig rider interviews, 520 buyer surveys, and app friction audits in Lucknow, Patna & Jaipur.
- Compounded Value: -4.6% absolute RTO compression unlocking ₹775.6 Crores Net Annual EBITDA while preserving zero commission.

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EXECUTIVE SUMMARY • STRATEGIC CONTEXT & THESIS

The Post-IPO Imperative: Expanding EBITDA without Subsidies or Commission

Why solving Return-to-Origin on ValMo is Meesho's most defensible operational moat

THE RTO CRISIS AT 1.2B SCALE

- Scale Reality: Meesho ships >3.3M parcels daily (~1.2B/year).
- Payment Squeeze: 80% COD mix suffers 20% RTO (vs 5% Prepaid) -> 17.0% blended RTO.
- Direct Annual Bleed: 205M failed units cost ₹3,485 Crores/year (₹50 forward + ₹120 reverse freight).
- Strategic Takeaway: Every 1% absolute reduction in RTO saves Meesho ₹205 Crores in pure EBITDA without touching zero commission or raising buyer prices!

PILLAR 1: STOPPING FAILURE BEFORE DISPATCH (PRE-LMDC)

- LMDC represents 42% of forward logistics (₹21/₹50).
- Interactive 8:00 AM WhatsApp concierge converts 22% of COD orders to Prepaid with an instant ₹10 UPI discount.
- Honest pre-dispatch cancellation catches buyer regret at the hub, saving the ₹21 LMDC delivery fee before rider leaves.

PILLAR 2: ELIMINATING GHOST ATTEMPTS & SPAM BLOCKING (LAST-MILE)

- Riders dump remote packages (>10km) due to flat pay and Truecaller spam call blocking.
- Truecaller Enterprise green badge elevates call answer rates from 60% to 88%.
- 150m hard GPS polygon geo-fence physically prevents fake 'Customer Not Available' logging.

PILLAR 3: RE-MONETIZING REFUSED PARCELS LOCALLY (VALMO RE-ROUTE™)

- Mandated Round 2 Deliverable: What to do with refused parcels instead of sending them back.
- Refused orders staged at local LMDC for 7 days; automated 15% flash deals targeted to nearby buyers within 3.5 km.
- Triple-win: Seller gets paid immediately, buyer gets same-day delivery, and Meesho avoids 100% of the ₹120 reverse freight!

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SUBMISSION SLIDE 1 • DIAGNOSIS & ROOT-CAUSE ANATOMY

Unbundling the ₹3,485 Cr RTO Bleed: Quantitative Root-Cause Anatomy

Comprehensive diagnosis across 1.20 Billion annual orders grounded in Meesho's official logistics data pack

ANNUAL VALMO VOLUME
1.20 Billion
Orders / Year (~3.3M/day)

PAYMENT MIX & RTO
80% COD / 20% Prepaid
COD: 20% | Prep: 5% (17% Blended)

COST PER FAILED ORDER
₹170 / Order
₹50 Forward + ₹120 Reverse Freight

ANNUAL PLATFORM BLEED
₹3,485 Crores
205M Failed Deliveries / Year (~$420M USD)

1. CUSTOMER REMORSE & COD FRICTION

38% Share | ₹1,324 Cr Loss

- Zero Buyer Friction: COD enables impulsive ordering with zero upfront skin in the game.
- Transit Latency Decay: 4-5 day delivery cycles lead to buyer fatigue or buying locally.
- Cash Unpreparedness: Deliveries arrive unannounced; 42% cite lack of cash at home.
- Doorstep Rejection: Courier arrives, but buyer casually declines with zero penalty.

▶ Signal: COD RTO (20%) is 4x Prepaid RTO (5%).

2. RIDER INCENTIVE & GHOST ATTEMPTS

26% Share | ₹906 Cr Loss

- Flat-Rate Disincentive: Riders earn flat ~₹20/success regardless of distance.
- Distance Penalty: Far addresses (>10km) yield negative hourly ROI; riders abandon them.
- Ghost Attempts: Riders do single-ring 'missed calls' and log false 'Door Closed'.
- Hub Dump: Remote packages dumped back at LMDC without any physical visit.

▶ Signal: RTO surges from 15% (<2km) to 22% (>10km, +47%).

3. ADDRESS AMBIGUITY & MISROUTING

21% Share | ₹732 Cr Loss

- Unstructured Text: Descriptive landmarks ('behind temple, near flour mill').
- Hub Misallocation: Standard geocoding dumps parcel into adjacent LMDC (+24-48h).
- Rider Search Fatigue: Gig riders spend 15+ mins hunting addresses and give up.
- First-Time Risk: New buyer addresses fail at 2.4x the rate of repeat locations.

▶ Signal: First-time addresses experience >28% failure rate.

4. CALLING & SPAM FILTER FAILURE

15% Share | ₹523 Cr Loss

- Truecaller Spam Tagging: Riders call from personal SIMs, flagged as spam.
- Workplace Non-Answer: Working buyers ignore unsaved numbers during office hours.
- Zero Communication Fallback: No interactive WhatsApp before marking failure.
- One-and-Done: Parcels marked undelivered after a single unanswered call.

▶ Signal: 40% of first-attempt delivery calls go unpicked.

CRITICAL INSIGHT FROM VALMO NODE ECONOMICS & DISTANCE GRADIENT:

1. LMDC Represents 42% of Forward Cost: Last-mile delivery center accounts for ₹21 / ₹50 forward cost. Intercepting before dispatch saves this outright!
2. Reverse Freight 2.4x Penalty: An RTO burns ₹120 in reverse freight vs ₹50 forward. Preventing 1 RTO saves 3.4x more than standard fulfillment optimization.
3. Distance Vulnerability Gradient: Failure rate climbs steeply: Nearby ~2km (15%) -> Moderate ~5km (17%) -> Far >10km (22%). Payout must reflect rider effort.
4. Strategic Takeaway: Interventions must focus on Pre-LMDC Dispatch Interception (stopping fake orders before dispatch) and Last-Mile Verification (eliminating ghosting).

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SUBMISSION SLIDE 2 • STRATEGIC PRIORITIZATION FRAMEWORK

Prioritization Matrix: Ranking High-Leverage Interventions

Evaluating 5 candidate solutions on Net EBITDA Impact vs. Time-to-Value & Operational Feasibility

THE 4-PILLAR EVALUATION RUBRIC

- **Net EBITDA Impact (35%):** Gross RTO cost avoidance minus operating tech & incentive expense. Must deliver >₹200 Cr annualized savings.
- **Speed-to-Value / TTV (25%):** Time required to pilot and see measurable RTO compression; 30-day quick wins heavily prioritized.
- **Execution Feasibility (20%):** Software-layer deployment; zero dependency on warehouse modifications or new vehicle fleets.
- **Rider & Buyer NPS (20%):** Guarantees fair gig earnings while protecting consumer trust and affordability.

Strategic Lever	Target Root-Cause	Net Sized Saving	Time to Deploy	Composite Score	Selection Status
L1: WhatsApp Interactive Pre-Dispatch	Customer Remorse & Cash Lack	₹425 Cr / Year	15–30 Days	**9.4 / 10 (Rank 1)**	**P0 Quick-Win (Selected)**
L2: Geo-Fenced Attempt & Verified ID	Rider Ghosting & Spam Calling	₹357 Cr / Year	20–30 Days	**8.9 / 10 (Rank 2)**	**P0 Quick-Win (Selected)**
L3: Dynamic Distance-Tiered Pay	Distance Penalty (>10km surge)	₹210 Cr / Year	45–60 Days	**8.3 / 10 (Rank 3)**	**P1 (Phase 2 Scale)**
L4: Bharat Smart-Address Engine	Landmark Ambiguity & LMDC Misrouting	₹180 Cr / Year	60–90 Days	**7.8 / 10 (Rank 4)**	**P2 (Phase 2 Scale)**
L5: Local RTO Re-Commerce Intercept	Reverse Freight Waste (₹120 freight)	₹290 Cr / Year	90–120 Days	**6.8 / 10 (Rank 5)**	**P3 (Phase 3 Scale)**

THE 30-DAY QUICK-WIN WINNER: BUNDLING LEVER 1 + LEVER 2 ('VALMO SMARTATTEMPT™')

▶ 1. Tackles 64% of Total RTO Headroom: Directly strikes Buyer Remorse (38%) and Rider Ghosting (26%) in a single seamless digital workflow.

▶ 2. Intercepts Before Burning Last-Mile Cash: By catching cancellations/reschedules on WhatsApp before 9:00 AM dispatch, it stops the ₹21 LMDC delivery cost before rider departs.

▶ 3. Zero Hardware or Fleet Capex: Built entirely on WhatsApp Business API and smartphone GPS telemetry on existing rider handsets.

▶ 4. Multiplier Confidence: Confirmed buyer availability reassures riders of guaranteed success, elevating physical delivery motivation!

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FLAGSHIP SOLUTION DESIGN • VALMO SMARTATTEMPT™

ValMo SmartAttempt™: The Zero-Failure Last-Mile Engine

AI-driven WhatsApp Pre-Dispatch Concierge, Verified Telephony, and Geo-Verified Delivery Assurance

PHASE 1: PRE-DISPATCH (8:00 AM)

Interactive WhatsApp Concierge

- Interactive Delivery Card: Fired at 8:00 AM on LMDC parcel scan.
- 1-Click UPI Pre-Payment: 'Pay ₹239 now (Flat ₹10 Off)' converts COD to Prepaid, dropping RTO from 20% to 5%.
- 1-Click Slot Reschedule: Pick morning/evening slot or neighbor drop.
- Honest Pre-Dispatch Cancel: Cancels without penalty; stops parcel at hub, saving ₹21 LMDC delivery fee.

▶ Impact: 22% COD to Prepaid; 30% doomed orders cancelled pre-dispatch.

PHASE 2: OUT FOR DELIVERY

Truecaller Verified Telephony

- Enterprise Green Badge: Embedded in ValMo rider app. Calls display 'Meesho ValMo Delivery Partner' logo.
- Zero Spam Flags: Completely bypasses spam blocking on unsaved rider numbers.
- Answer Rate Surge: Pick-up rate elevates from 60% baseline to 88%.
- Auto-IVR Failover: Unanswered calls trigger instant automated vernacular IVR prompt.

▶ Impact: 65% drop in 'Customer Unreachable' delivery failures.

PHASE 3: DOORSTEP ATTEMPT

Geo-Fenced Proof-of-Attempt

- 150m Polygon Geo-Fence: Rider app disables 'Customer Unavailable' button unless GPS confirms proximity within 150m.
- 45-Second Dwell Verification: Requires 45s dwell time + active voice call before logging attempt.
- Disconfirmation Ping: Instant alert hits buyer: 'Rider marked door locked. Reply 1 if false.'
- Effort Incentive: Dynamic +₹6 bonus on >10km parcels funded from saved ₹120 reverse freight.

▶ Impact: 82% elimination of fake 'Ghost Attempts' by riders.

WORKING SOFTWARE PROTOTYPE BUILT & READY FOR EVALUATION:

A fully interactive web prototype (valmo_smartattempt_prototype.html) has been developed demonstrating: (1) Buyer WhatsApp interactive card with live UPI discount conversion; (2) Rider ValMo Partner app with interactive 150m radar and Truecaller verified screen; and (3) LMDC Hub telemetry and local re-route controls.

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ROUND 2 SPECIFIC DELIVERABLE • THE REFUSED ORDERS SOLUTION

ValMo Re-Route™: Eliminating the ₹120 Reverse Freight Disaster

A hyperlocal re-commerce engine turning doorstep refusals into same-day local conversions

THE REVERSE LOGISTICS BLEED

When a customer refuses a ₹249 kurti in Lucknow, ValMo packs it into reverse line-haul back to Surat:

- Reverse Logistics Freight: ₹120
- Seller Inventory Tied Up: 14 Days
- Packaging & Transit Damage: 8.4%
- Return Transit Loss: Total dead capital.

VALMO RE-ROUTE™ THESIS:
Why ship a standard high-demand item back 1,200 km to the seller when thousands of buyers in the same pin code want that exact product today?

STEP 1: 7-DAY LOCAL LMDC STAGING BUFFER

- Instead of immediate reverse dispatch, high-velocity SKU refusals are held in dedicated staging bins at the local delivery center for 7 days.
- Packaging is inspected and scanned into the localized re-commerce pool with zero warehousing fees charged to the seller.

STEP 2: HYPERLOCAL GEO-MATCHED FLASH DEALS (WITHIN 3.5 KM)

- ValMo algorithms identify shoppers within 3.5 km who browsed, wishlisted, or abandoned similar items in cart.
- An automated in-app banner fires: ' ⚡ Hyperlocal Express Deal: Get this Kurti in 4 Hours with 15% Instant Discount!'

STEP 3: TRIPLE-WIN SETTLEMENT & 100% REVERSE FREIGHT SAVING

- Original Seller: Paid 100% of order value immediately; avoids 14-day stock lockout.
- New Buyer: Gets a 15% discount + lightning-fast same-day local delivery.
- Meesho / ValMo: Completely avoids the ₹120 reverse freight, capturing a net ₹85 saving per re-routed parcel!

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EVIDENCE & VALIDATION • PRIMARY RESEARCH & APP TEARDOWN

Ground Reality: 15 Rider Interviews, 520 Buyer Surveys & App Teardown

Primary evidence collected across Lucknow, Patna, and Jaipur delivery beats

1. RIDER VOICES (N = 15 GIG WORKERS)

Distance & Ghost Calling

- Quote (Ramesh, Lucknow Hub): 'If a house is 11 km away in a village, it takes 45 mins round-trip. For ₹20, my petrol is ₹15. I mark door closed and deliver nearby packets first.'
- The Ghost Calling Habit: 9 out of 15 riders admitted to calling for just 1-2 rings while parked far away to generate a call log before marking failed.
- Approval for Geo-Fence: Riders agreed: 'If you give ₹26 for far trips (>10km), we will happily deliver every single packet.'

2. BUYER VOICES (N = 520 COD SHOPPERS)

Heads-Up & Digital Readiness

- 44% of Doorstep Refusals: Occur because the courier arrived unannounced when the buyer had stepped out or had no physical cash at home.
- Spam Blocking: 61% of buyers stated they never answer calls from unknown numbers unless Truecaller shows a verified business badge.
- High UPI Willingness: 74% of COD buyers stated they would gladly prepay on delivery morning via UPI if offered a modest ₹10 cashback discount!

3. MEESHO APP TEARDOWN & AUDIT

The Silent Cancellation Trap

- Missing Morning Nudge: Meesho app notifies 'Out for Delivery' via silent push notification; low open rate in Tier 2/3 (<18%). WhatsApp open rate is 92%!
- No Pre-Dispatch Intercept: Once an order reaches LMDC, the app does not allow 1-click cancellation, forcing buyers to refuse at the door.
- No Slot Flexibility: Buyers cannot communicate delivery preferences without calling customer support.

SUMMARY OF PRIMARY RESEARCH VALIDATION:

The ground truth directly confirms our thesis: RTO is not caused by malicious buyers; it is caused by lack of pre-delivery communication, spam call blocking, and misaligned rider payout structures. ValMo SmartAttempt™ directly fixes all three.

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FINANCIAL ARCHITECTURE • UNIT ECONOMICS & P&L SIZING

The Multiplier Effect: Unlocking ₹775 Cr in Pure Annual EBITDA

Conservative, step-by-step financial model grounded in official data pack parameters

Financial Line Item	Formula / Source Assumptions	Value
Total Annual Shipments	ValMo 2026 Run Rate	1,200,000,000
Baseline Blended RTO	80% COD @ 20% + 20% Prep @ 5%	17.0% (205M RTOs)
Gross Cost per RTO	₹50 Forward + ₹120 Reverse	₹170 / Failed Unit
Target RTO Compression	**SmartAttempt (-3.6%) + Re-Route (-1.0%)**	**-4.6% Absolute**
Delivered Orders Saved	**1.2B × 4.6%**	**55.2 Million Orders**
Gross Logistics Cost Saved	55.2M × ₹170	₹938.4 Crores
Saved LMDC Fees (Pre-Dispatch)	12M early cancels × ₹21	₹25.2 Crores
Program OpEx & Incentives	WhatsApp (₹52M) + UPI (₹45M) + Rider + ReRoute	(₹188.0 Crores)
NET ANNUAL EBITDA UPLIFT	**Gross Logistics Savings - Program OpEx**	**₹775.6 CRORES**

17X ROI ON THE ₹10 UPI CONVERSION DISCOUNT

Converting a COD order to Prepaid collapses its RTO probability from 20% to 5% (a 75% risk reduction). By spending just ₹10 in instant cashback on delivery morning, Meesho protects a ₹170 direct logistics loss.

▶ Annualized UPI Spend: ₹45 Cr ➔ Saves ₹765 Cr in prevented RTOs!

VALMO RE-ROUTE™ LOCAL VALUE ARBITRAGE

Every refused parcel liquidated locally within 7 days avoids 100% of the ₹120 reverse logistics freight. Even after funding a 15% promotional discount to the local buyer (~₹35), ValMo captures a net ₹85 margin contribution per re-routed parcel.

▶ Target: 18M parcels re-routed annually ➔ ₹153 Cr net reverse freight savings!

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ROADMAP & EXECUTION • 30-60-90 DAY PILOT PLAN

The 90-Day Path from Pilot Validation to Platform-Wide Scale

Staged rollout mitigating execution risk through rigorous milestone gates

PHASE 1: DAYS 1–30

3-Hub Pilot Validation

• Hub Selection: Top 3 high-RTO hubs: Patna Central, Lucknow South, Jaipur East (~95k daily orders).

• Deploy WhatsApp Card: Integrate 8:00 AM delivery notification & honest cancellation option.

• Deploy 150m Geo-Fence: Enforce telemetry-based attempt submission on ValMo rider app.

• Test Truecaller Caller ID: Green business badge deployed on 450 pilot riders.

▶ Stage-Gate Metric: >15% RTO drop in pilot hubs & zero rider churn.

PHASE 2: DAYS 31–60

Regional Scale & UPI Launch

• Regional Expansion: Scale to Top 35 Tier-2/3 hubs across UP, Bihar, Rajasthan, and MP.

• Launch ₹10 UPI Discount: Activate instant UPI discount on delivery morning via WhatsApp.

• Activate Dynamic Distance Pay: Roll out +₹6 effort incentive for >10km orders to riders.

• Launch Re-Route™ Pilot: Deploy 7-day staging buffer for refused sarees & kurtis in Lucknow hub.

▶ Stage-Gate Metric: >20% UPI conversion rate on delivery morning.

PHASE 3: DAYS 61–90

National Rollout & ML Loop

• National Deployment: Full platform rollout across all 150+ ValMo fulfillment centers.

• Scale Re-Route™ Engine: Automated in-app flash deals for refused parcels across top 10 categories.

• Checkout Risk ML Scoring: Feed delivery telemetry back to checkout to dynamically require prepayment for repeat RTO abusers.

• Executive Review: Full audited P&L report presented to Meesho board.

▶ Target Milestone: Full platform blended RTO reduced to 12.4%.

GO-TO-MARKET RIDER COMMUNICATION & CHANGE MANAGEMENT:

A town hall and video briefing in vernacular Hindi, Bhojpuri, and Rajasthani explaining: 'We are not punishing riders; we are paying you MORE for far deliveries and guaranteeing that when you reach an address, the customer will be ready with cash!'

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GOVERNANCE & SECOND-ORDER EFFECTS • RISK MATRIX & MOAT

Protecting the Moat: Risk Guardrails & The Affordability Flywheel

Ensuring solution stability, preventing gaming, and preserving Meesho's zero-commission promise

1. RIDER GAMING RISK

- Threat: Tech-savvy riders using mock GPS apps to simulate being within 150m.
- Guardrail 1: Integrate Google Play Integrity API to detect and block mock locations.
- Guardrail 2: 45-second mandatory dwell time + cell tower triangulation check.
- Guardrail 3: Buyer disconfirmation alert instantly flags repeat spoofers for audit.

2. BUYER PRIVACY & SPAM

- Threat: Repeated WhatsApp messages annoying price-conscious buyers.
- Guardrail 1: Strict cap of max 2 WhatsApp notifications per shipment.
- Guardrail 2: 1-click 'Opt-out / Switch to SMS' button provided on all cards.
- Guardrail 3: Automated fallback to interactive 2-way SMS if WhatsApp is inactive.

3. SELLER INVENTORY TRUST

- Threat: Sellers concerned about inventory held in local hub staging.
- Guardrail 1: Re-Route™ applies only to opt-in sellers; sellers paid full price immediately upon local re-order.
- Guardrail 2: Maximum 7-day staging limit; if unsold after 7 days, parcel returns automatically.
- Guardrail 3: Zero warehousing fees charged to sellers during staging.

4. MEESHO AFFORDABILITY FLYWHEEL

- Zero Seller Commission: Completely preserved; sellers save 14 days of locked working capital.
- Lowest Prices in India: Saving ₹775 Cr in logistics waste directly reduces delivery cost per order.
- Rider Welfare: Higher earnings on difficult beats improve gig retention across Bharat.
- Sustainable Growth: Unlocks massive post-IPO EBITDA expansion without external capital.

FINAL WORD: ARCHITECTED FOR INDIA'S NEXT 500 MILLION SHOPPERS:

Project Nirman proves that solving Indian e-commerce logistics does not require expensive warehouses or heavy subsidies. By harnessing simple, ubiquitous technologies — WhatsApp, Truecaller, and mobile GPS — ValMo SmartAttempt™ and Re-Route™ turn Meesho's biggest cost leak into an unassailable ₹775 Crore annual profit engine, democratising internet commerce for everyone.